



SetuDocs AI

Paperwork, sorted.

Document intelligence for families, small businesses, CAs and CSC operators

Technical Project Documentation

Pragyan 2K26 · National-Level Hackathon · Track 4: Entrepreneurship & Future of Management

Team SetuDocs AI

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At a glance

What SetuDocs AI is, in one page

SetuDocs AI turns scanned and photographed paperwork, in Marathi or English, into a searchable record that **warns you before deadlines** and **shows the schemes you qualify for**.

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roles: Admin, Officer,
Citizen

82

automated tests, all
passing

8

schemes and
registrations matched

21

document types
recognised

Who it is for

Small business owners and families in tier-2 and tier-3 towns, and the CAs and CSC operators who handle papers for many clients.

What it does

Reads documents with OCR, understands them with AI, then warns about deadlines, matches schemes and shows the impact.

Why it is different

It does not just store files. It understands them, checks whether they can be trusted, and tells you what to do next.

Where it stands

A working prototype, built on our earlier project GovDocs AI, with every capability labelled by its real state in this document.

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The problem

Paperwork costs people most when they notice it too late

Individuals, families and small businesses lose time, money and opportunities because their paperwork is scattered, scanned, in mixed languages and impossible to search or track.

Can't find it

Licences and certificates sit in bags, folders, phone galleries and chats. One copy can take an hour to find.

Can't search it

A photo of a document is just a picture. No search can read the text inside it.

Misses the deadline

Renewals and expiry dates lapse unnoticed. The penalty is often the first warning.

Misses the scheme

Few people link their documents to schemes they qualify for, or know what they still lack.

Two languages

Marathi text with English numbers and codes on one page breaks single-language tools.

No trust signal

An upload may be outdated or contradicted by a newer document, and nothing says so.

Who is affected

Primary users: shop owners, vendors, micro-enterprises and families. **Multiplier users:** CAs and Common Service Centre operators, who each serve dozens of clients. **Operators:** an administrator who runs the system.

3 The solution

Upload once. SetuDocs does the rest.

Setu means *bridge* in Marathi. SetuDocs AI bridges people and the paperwork systems they deal with, without changing how they work.

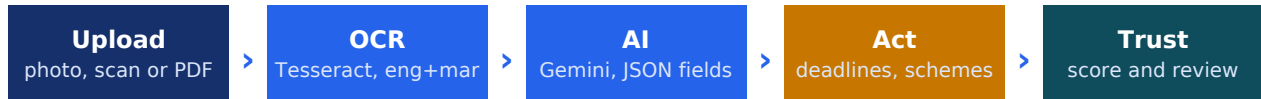


Fig. 1. From upload to action. Every step is written to the audit log.

1 Read

Tesseract OCR turns the image into text, reading English and Marathi on the same page.

2 Understand

Google Gemini returns structured fields: title, summary, department, category, keywords, confidence and dates.

3 Act

Deadline Guard warns early, Scheme Matcher shows likely schemes and missing documents, Impact shows the result.

4 Trust

Nothing is trusted by default. Each document gets a trust score and status, and a person makes the final call.

AI output is a suggestion, not a decision. The OCR text stays visible next to every AI field, and deadline and scheme logic is rule-based so every result can be explained.

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Features

What is built, and its real state

Capture and understand

Secure upload	Type, 10 MB limit, file-signature check, random server file names	● Working
Document check	Rejects photos that are not documents	● Working
OCR, English	Tesseract	● Working
OCR, Marathi	eng+mar; tested in GovDocs AI, needs the mar pack installed	● Tested
AI extraction	Gemini fields plus dates, English or Marathi summary	● Tested
Smart Search	9 fields, typo-tolerant fallback, filters, sorting	● Tested

Trust and control

Trust & Verification	Trust score, status, claims, contradictions, review	● Working
Review workflow	Approve, reject, send back, archive	● Working
Roles	Admin, Officer, Citizen with server-side checks	● Working
Citizen portal	Self-registration, Citizen ID, own documents	● Working
Audit log	Who did what, and when	● Working

Act on it

Deadline Guard	Finds dates, groups them by urgency	● Tested
Scheme Matcher	8 schemes and registrations, readiness, missing documents	● Tested
Impact	Documents, deadlines, time saved (estimate until measured)	● Tested
Marathi and English interface	Whole interface switches language	● Working

Also		
Analytics, settings, PDF export	Admin oversight and one-page summary	● Working
Recovery Center	Data-recovery demonstration	● Simulation
Bilingual synonym search	In the code but not connected	● Built, not connected
Marathi-script search, OCR accuracy %, load	No test run yet	● Not tested

Working: implemented and checked by hand. **Tested:** covered by automated tests (Marathi OCR: tested in GovDocs AI). **Simulation:** a demonstration only. **Not tested:** not claimed.

5 How it is built

Architecture and technology

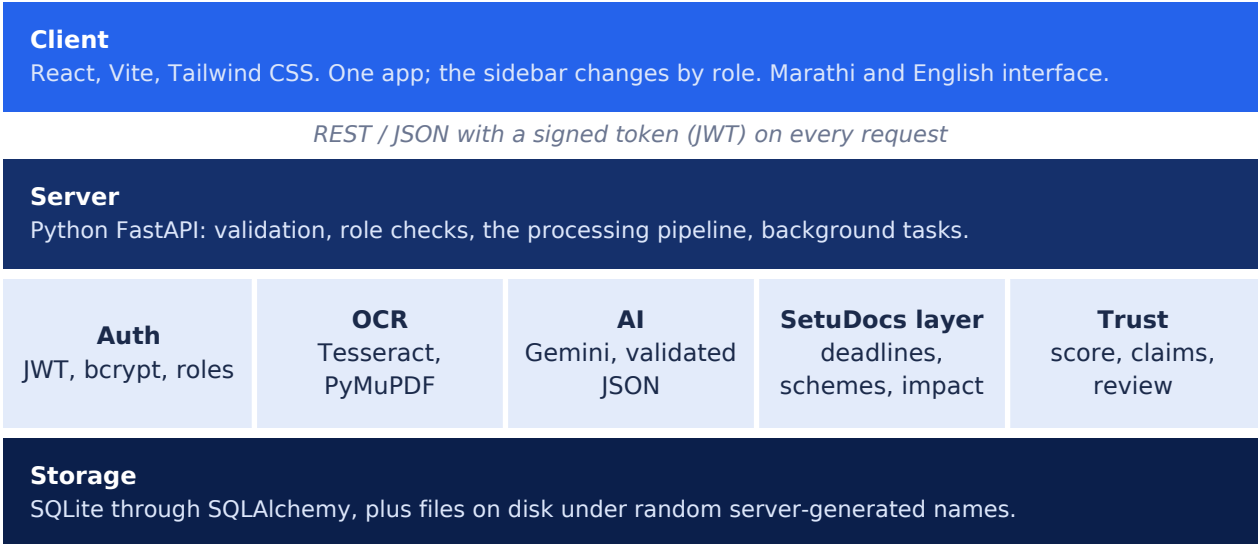


Fig. 2. Each service sits in its own module, so OCR, AI or the database can be replaced without rebuilding the rest.

Technology

Part	Technology	Part	Technology
Frontend	React 18, Vite, Tailwind, Recharts	AI	Google Gemini Flash-Lite
Backend	Python, FastAPI, Pydantic	Search	rapidfuzz (typos)
Database	SQLite + SQLAlchemy	Security	JWT, bcrypt
OCR	Tesseract 5 (eng+mar), PyMuPDF	Testing	pytest, 82 tests

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Who can do what

Three roles, checked on the server

Action	Admin	Officer	Citizen
Log in, profile, upload documents	✓	✓	✓
Register a new account	-	-	✓ self-service
See documents	All	All	Own only
Deadline Guard and Impact	All	All	Own
Scheme Matcher	Own profile	Own profile	Own profile
Smart Search and Review Queue	✓	✓	-
Trust & Verification decisions	✓	✓	Submit for review
Archive	✓	-	-
Analytics, audit log, settings, Recovery Center	✓	-	-

The role inside the signed token is checked before any request runs. A Citizen asking for someone else's document is told it does not exist.

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How a document is processed

From a phone photo to an alert

1	Check	Extension (PDF, JPG, PNG), size up to 10 MB, and the file's real signature are checked. A quick test rejects photos that are not documents.
2	Store	The file is saved under a random server-generated name with a SHA-256 hash, so names cannot clash or be abused.
3	Read	OCR runs in the background. PDFs are rendered page by page; English and Marathi are read in one pass.
4	Understand	Gemini returns validated JSON. A bad reply is stored as a failure, never saved half-complete; temporary errors are retried.
5	Enrich	Document types are detected (identity numbers are never stored), deadlines are found, and a trust score is calculated.
6	Find	Smart Search checks 9 fields; if nothing matches, a fuzzy pass catches typos.
7	Track	Uploaded → OCR → AI → Pending Review → Approved, Rejected, Needs Correction or Archived. Failed steps can be re-run.

Deadline Guard

Dates come from two sources and are merged: the dates Gemini labels, and a rule-based reader of the OCR text. A date only counts when a cue such as *valid till*, *expiry* or *due date* (or the Marathi equivalent) sits next to it; issue and birth dates are ignored. Users can add, edit or dismiss deadlines, and their changes survive a re-scan.

Overdue date has passed	Critical 0 to 7 days	Soon 8 to 30 days	Upcoming more than 30 days
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Scheme Matcher

Covers Udyam Registration, PMEGP, PM MUDRA Yojana, Stand-Up India, CGTMSE, PM SVANidhi, PM Vishwakarma and PM-KISAN. Each is a set of rules on the signed-in user's business profile. A scheme is **likely** when every condition holds, **possible** when some profile details are missing, and hidden when a condition fails. Each match shows a readiness percentage, the required documents (ticked when already uploaded), what is missing, and the official link.

Scheme rules are guidance, not an eligibility decision. They are hand-curated and must be checked against official sources.

Impact

Shows documents digitised, deadlines by urgency, scheme matches, the document mix and a 14-day timeline. Time saved = documents × (time by hand – time with SetuDocs). It uses assumed times (5 minutes by hand, 20 seconds with SetuDocs) and is labelled **Estimate** until there are three real timing trials for each method; then it switches to **Measured**.

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Trust and security

Nothing is trusted by default

Trust score

0 to 100. Raised by an official source and supporting documents; lowered by conflicts and suspicious wording.

Status

Verified, Corroborated, Needs review, Flagged, or Outdated when a newer official document replaces it.

Claims

Key facts (dates, amounts, schemes, reference numbers) are pulled out and compared across documents.

Human decision

Staff resolve the Review Queue. Citizens can submit their own document. Every step is kept in its history.

Security controls

Area	Control
Passwords	Stored only as bcrypt hashes
Sessions	Signed tokens, 30-minute expiry, checked on every request
Access	Roles checked on the server; Citizens see only their own documents
Login	5 wrong passwords in 5 minutes locks the account for 5 minutes
Uploads	Type, size and signature checks; random file names
Secrets	Keys only in environment files, never sent to the browser; the app refuses to start in production with default secrets
Records	Audit log of every important action

Recovery Center (simulation)

Demonstrates how data could be protected if the main store failed: a checksummed snapshot, a simulated outage that queues new writes, and a recovery run that restores records and reports what was in flight. It is a demonstration, not production disaster recovery.

Testing and results

What is proven, and what is not

Test suite	Tests	Covers
Smart Search	24	Every field, word order, typos, filters, sorting, paging
SetuDocs layer	46	Document types, deadlines in English and Marathi, urgency groups, re-scans, scheme matching, Impact estimate vs measured
AI service	12	JSON parsing, missing fields, retries, missing key, confidence limits
Total	82	All passing, 6 October 2026

From GovDocs AI testing (same OCR and AI modules)

A real Maharashtra Government Resolution was read in one pass with English and Marathi together: **2,622 characters**, correctly classified at **95% AI confidence**.

Checked by hand, not automated: logins and roles, upload checks, Trust & Verification, the Recovery Center and the interface. **Not tested, so not claimed:** Marathi-script search, OCR accuracy as a percentage, heavy load, and measured time saved.

Business and impact

Feasible today, ready to grow

Individuals and small businesses

Free tier for storage and search; a low monthly plan for alerts and scheme matching.

CAs and CSC centres

A multi-client plan. One customer brings dozens of users: the fastest route to revenue.

Feasible now

Working end to end, built on free open tools; only the AI step uses a paid API.

Ready to scale

PostgreSQL is a configuration change; OCR and AI can move to workers; new schemes are added as data.

Prices are hypotheses we will test with real users.

Sustainable Development Goals

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Decent work and economic growth

Fewer missed renewals and penalties

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Industry, innovation and infrastructure

Affordable digital tools for small firms

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Peace, justice and strong institutions

Easier access to services; documents with a trust status

Limitations and next steps

Honest limits, and what comes next

Limitation	What it means
Data residency	Text goes to Google's Gemini API, hosted outside India. The AI sits in one module, so an India-hosted model can replace it.
Internet needed	AI analysis depends on an outside service.
SQLite	One writer at a time, no redundancy: fine for a prototype.
AI can be wrong	Confidence scores, visible OCR text and human review reduce this, but do not remove it.
Scan quality	Blurry or tilted scans lower OCR accuracy, Marathi more so.
Prototype only	No hosting, HTTPS, backups or monitoring yet.

Next steps (not built yet)

- **Ask your documents:** ask a question in Marathi or English and get an answer from your own papers.
- **Document health check:** what a business has, what is expiring and what is missing.
- **CA share pack:** one indexed bundle of chosen documents for a CA or office.
- **Reminders** by SMS or WhatsApp; **bilingual synonym search** connected after testing.
- **Production:** India-hosted AI, PostgreSQL, cloud storage.

Team and origins

Built on GovDocs AI

SetuDocs AI grows out of GovDocs AI, our earlier document system for government offices (Smart Kopargaon Hackathon), which provided upload, Marathi and English OCR, AI metadata, search, review, audit and analytics. SetuDocs adds the Citizen role and portal, Deadline Guard, Scheme Matcher, Impact, Trust & Verification, typo-tolerant search, the document check, the Marathi interface, the Recovery Center simulation and a new brand.

Member	Role
Parth Pawar	Team Leader, Technical Lead & Full-Stack Developer
Tanmay Shinde	Design & Presentation
Ishwari Mroe	Presenter
Kurshna Gaikwad	Team Member

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Appendix: data model

Main tables

Table	Holds
users	Username, password hash, role, name, Citizen ID
documents	File details, status, OCR text, AI fields, review fields, file hash
deadlines	Label, kind (expiry, renewal, due), date, source, status
business_profiles	Entity type, stage, sector, state, gender, social category
document_insights	Detected document types
timing_trials	Manual and SetuDocs timings for Impact
document_verifications	Trust score, band, status, source details, review
document_claims	Facts pulled from documents
claim_contradictions	Conflicts between documents and how they were resolved
verification_events	History of each verification
audit_logs	User, action, document, time
app_settings	AI confidence threshold

B

Appendix: main API routes

Full, live documentation is at </docs> while the server runs

Area	Routes	Who
Auth	/api/auth/register , login , logout , me , change-password	Public / any
Documents	/api/documents/upload , {id} , download , export PDF ; review , edit , delete	Any (own) / staff
Citizen	/api/citizen/dashboard , documents	Citizen
Deadlines	/api/deadlines (list, add, edit, delete, rescan)	Any (own data)
Schemes	/api/schemes/matches , profile	Any
Impact	/api/impact , trials	Any (own data)
Verification	/api/verification/review-queue , analyze , decision , claims , history	Staff; Citizens submit
Admin	/api/analytics , audit-logs , settings , recovery	Admin



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